FNCE30007 Derivative Securities

Tutorial 7: BSM Model Part 1

(week commencing 21/09/2026)

QUESTION 1

Let X be a normally distributed random variable with mean 100 and variance 100.
(a) What is the probability that X will take on a value less than 85.
(b) What is the probability that X will take on a value greater than 90.
(c) What is the probability that X will take on a value less than 85 or more than 90.

QUESTION 2

Let Z be a standard normal random variable.
(a) What is the probability that Z will take on a value between -1 and 1.
(b) What is the probability that Z will take on a value between -2 and 2.
(c) Comment on your answers in parts (a) and (b).

QUESTION 3

The current price of a stock is $50. Assume the expected return on the stock is 18 % p.a. and its volatility is 30% p.a.. In the Black-Scholes framework, what is the probability distribution of the stock price in:
(a) one year time?
(b) two years' time?

QUESTION 4

Consider a non-dividend paying stock when the stock price is $10.00, the risk-free interest rate is 6% p.a. and the volatility is 40% p.a..

a) use the Black-Scholes call option formula to calculate the current price of a 2-year European call option (strike $10.00) on the stock
b) use the Black-Scholes put option formula to calculate the current price of a 2 year European put option (strike $10.00) on the stock
c) use the simulator to check your answers.

QUESTION 5

Reconsider the European call option in question 4(a). Recalculate the price of the option using:

a) a one-period binomial model.
b) a ten-period binomial model using the simulator
c) what do your answers in parts (a) and (b) suggest about the relationship between the binomial and Black-Scholes models?

QUESTION 6 - PLEASE ALSO REFER TO THE CASE STUDY MATERIAL POSTED FOR LECTURE 6 (Week 7 Case Study, and note this was originally part of Tutorial 6 questions)

After reading the Virgin case, complete the following questions before the tutorial.

The Effect of Increasing the Number of Periods

Now focus on Option C: Jan 2027 Brent Call, which has the longest time to maturity.

Revalue the option using: N = 2, 4, 8, 16

6.1 Calculation - USE MARKET DATA PREVIOUSLY PROVIDED BUT NOTE THAT THE NUMBER OF DAYS IS INCORRECT (Days to expiry is 84 not 83)

For each value of N, recalculate $\Delta t = T/N$ and the corresponding u, d and p. Use the simulator to construct the larger binomial trees where appropriate and compare the estimates in this table.

Number of Periods, N	Binomial Value	Observed Market Price	Difference: Model - Market	Absolute % Pricing Error
2		7.72		
4		7.72		
8		7.72		
16		7.72		

6.2 Interpretation

- How does the estimated option value change as the number of periods increases?
- Does the binomial estimate appear to become more stable as the tree becomes finer?
- How does the pricing error change as N increases?
- What does the pattern suggest about convergence of the binomial estimate as the tree is refined?

Data note: Option-chain figures are based on Barchart market data observed on 2 September 2026. Prices are quoted in USD per barrel. Barchart reports an option point value of USD 1,000.